SOC Log Analysis Report – Failed Login Detection

Introduction

Security Operations Centers (SOC) are responsible for monitoring, detecting, analyzing, and responding to cybersecurity incidents. This project focuses on log analysis, threat intelligence integration, and incident response using SIEM tools.

Objective

- Detect suspicious activities using log analysis
- Correlate logs to identify attack patterns
- Enrich alerts with threat intelligence
- Perform incident escalation and reporting

Environment Setup

Tools Used:

- Kibana
- Elasticsearch
- Windows System

Description:

Kibana was used to visualize and analyze log data. Windows system logs were generated to simulate failed login attempts.

Step-by-Step Procedure

Step 1: Access Kibana Dashboard
Open browser
Navigate to:
http://localhost:5601
Login using credentials

Step 2: Open Discover Tab
Navigate to Discover
Select index pattern (e.g., logs-*)

Step 3: Verify Log Data
Enter search query: log*
Confirm logs are visible
Step 4: Generate Failed Login Logs
Open Command Prompt:
Win + R → cmd
Run: net user test wrong pass

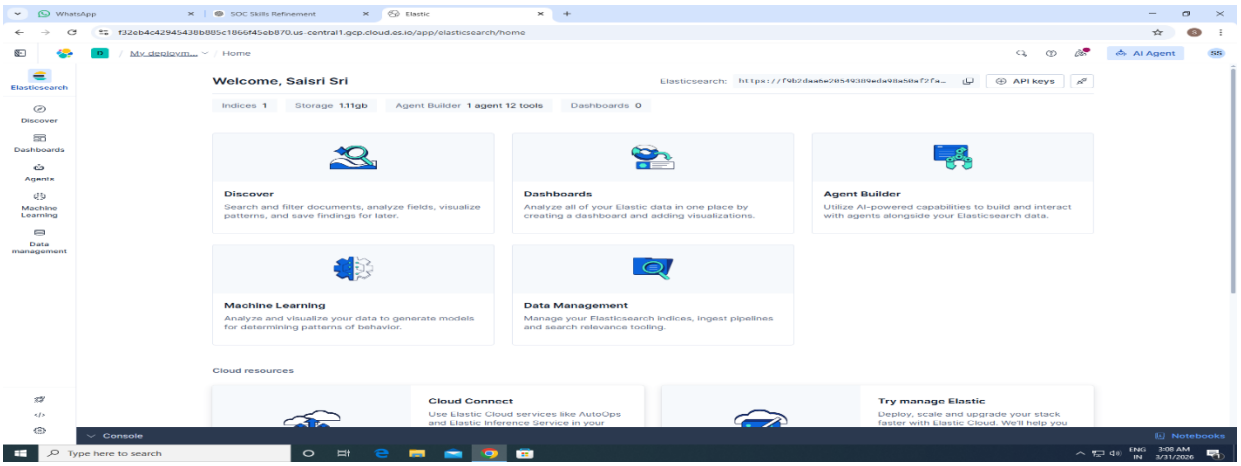
Step 5:
Search Failed Login Events
In Kibana Discover:
Event code:4625

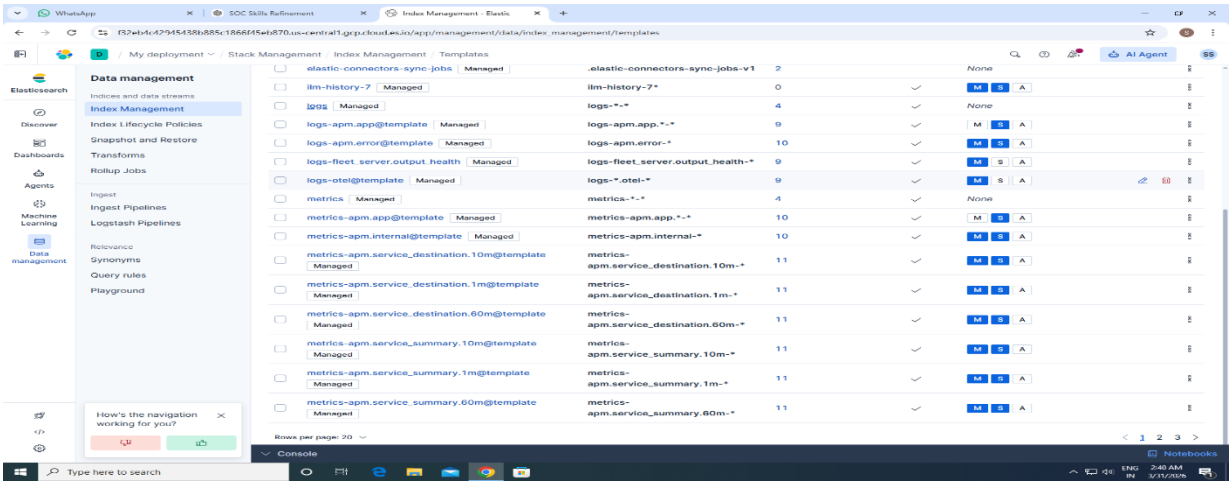
Step 6: Analyze Logs
Observe:
Multiple failed attempts
Same username
Close timestamps

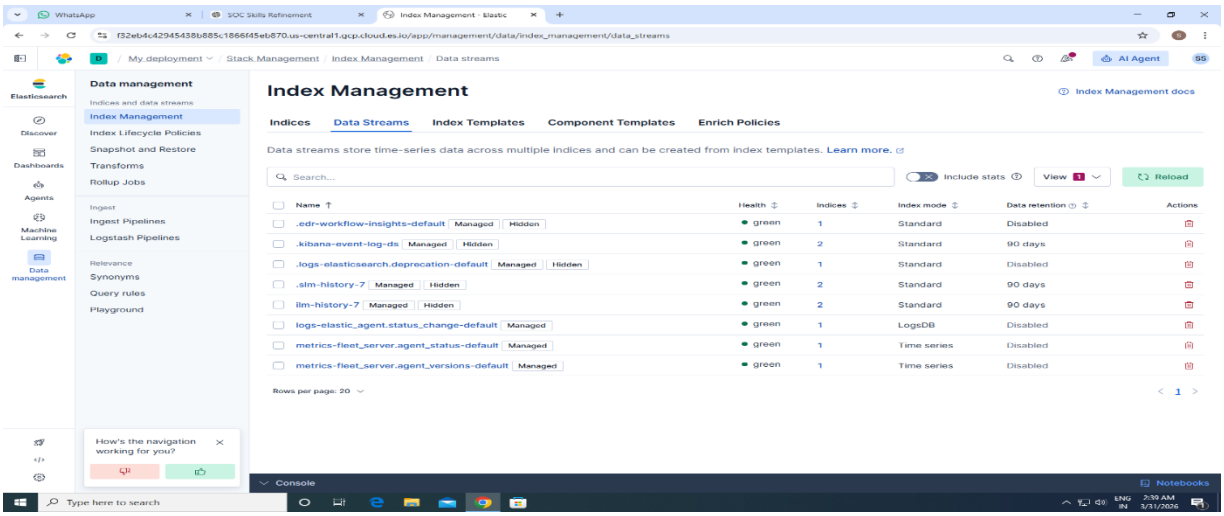
Step 7: View Event Details
Click on a log entry
Analyze:
Event ID
Username
Timestamp

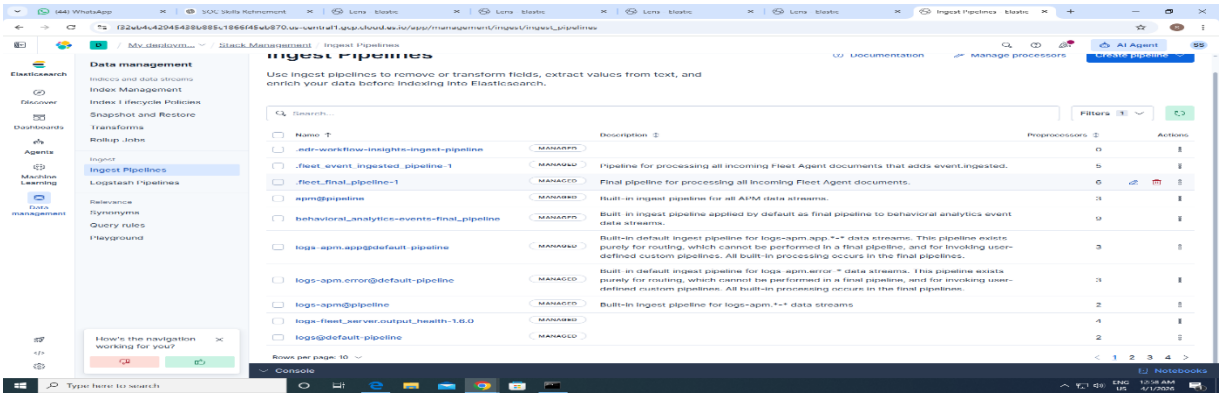
Step 8: Threat Intelligence Check
Copy IP address from logs
Check in Virus Total
Record findings

Failed Login Events
Kibana Discover with 4625 logs



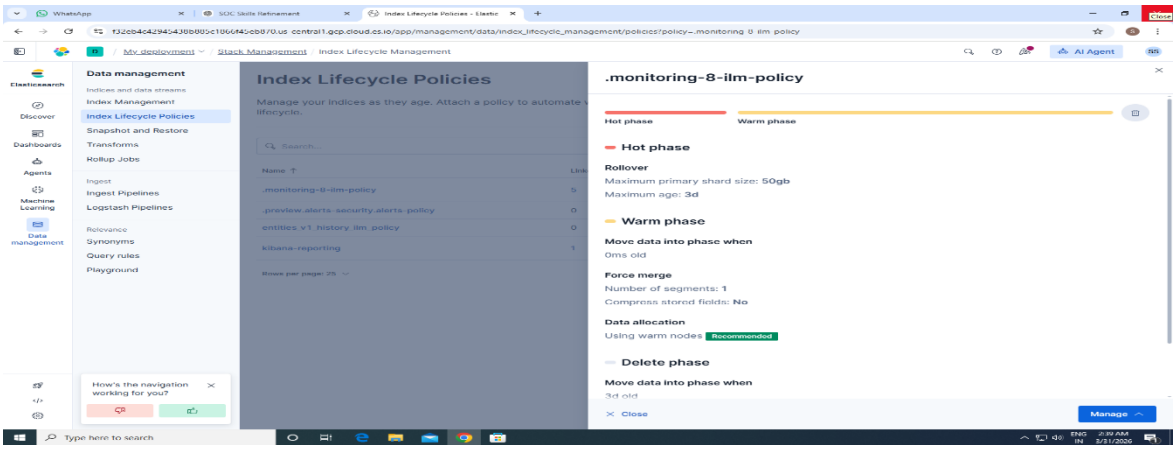


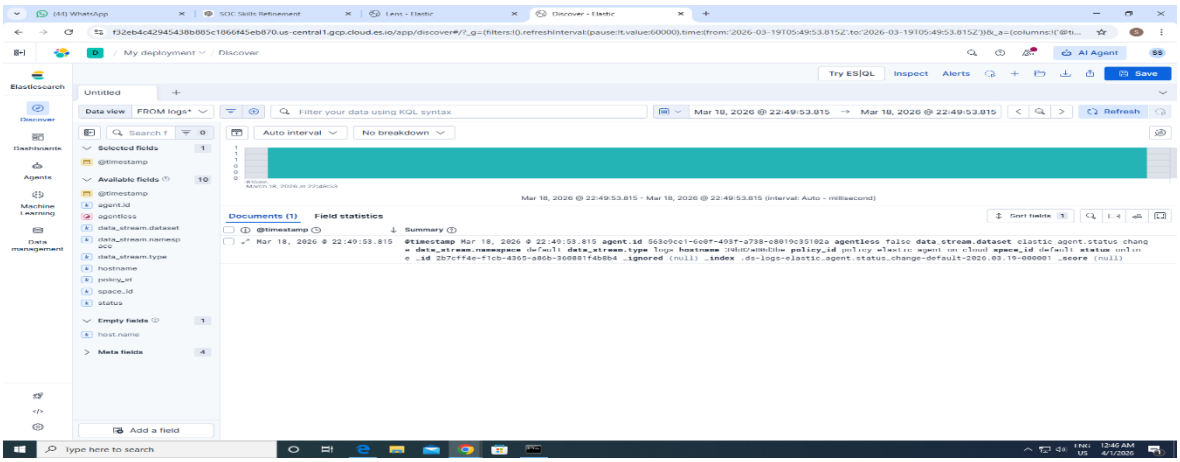


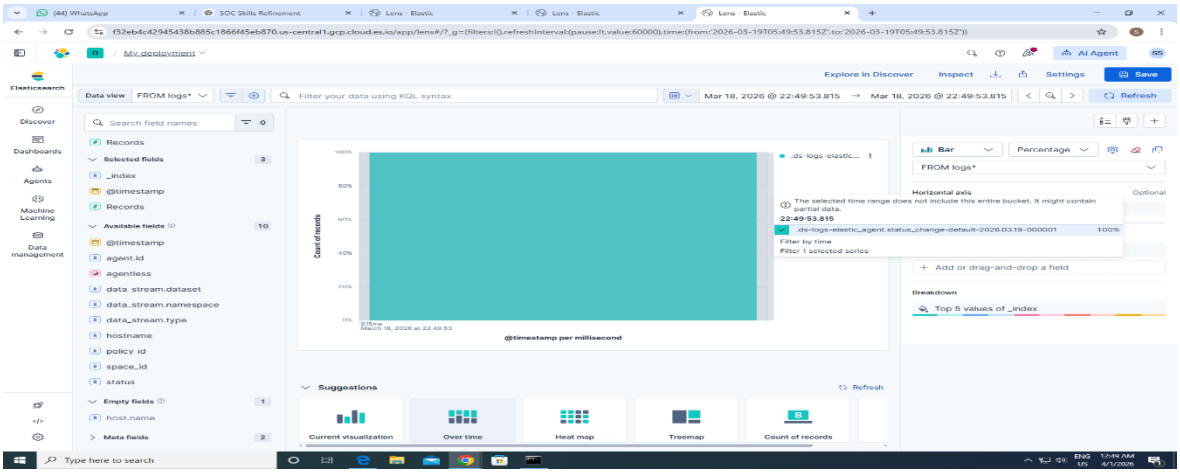


Explanation: This screenshot shows multiple failed login attempts detected in Kibana

Event Details:

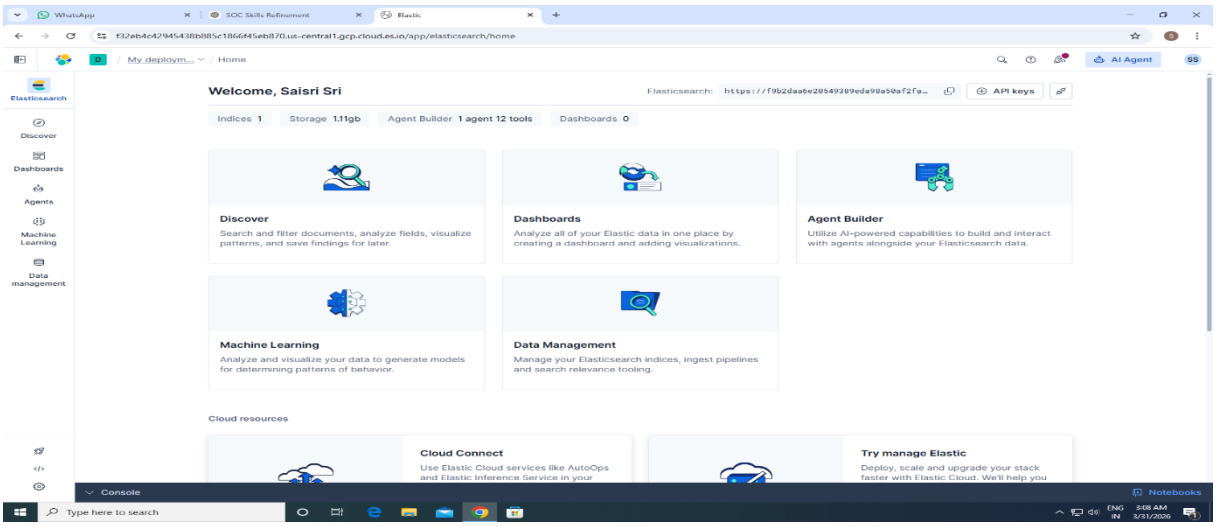






Explanation:
This screenshot displays detailed information of a failed login event.

Kibana Dashboard



Log Analysis Table

Timestamp	Event ID	Username	Observation
Kibana	4625	test	Failed login detected
ELK	4625	test	Failed login detected

Analysis & Correlation

Multiple failed login attempts were observed within a short time period. This pattern indicates a possible brute-force attack where an attacker tries multiple passwords to gain unauthorized access.

Threat Intelligence Findings

The IP address associated with the login attempts was analyzed using Virus Total. No strong malicious reputation was identified, indicating either benign activity or unknown threat.

Findings

Repeated failed login attempts detected
Same user targeted multiple times
Pattern indicates suspicious activity

Recommendations

- Implement account lockout policies
- Use strong passwords
- Enable multi-factor authentication
- Monitor login activity regularly

Conclusion

The analysis suggests a potential brute-force attack based on repeated failed login attempts. Continuous monitoring and preventive controls are recommended